

## Investigation of Long Short-Term Memorys to Temperature Prediction for Permanent Magnet Synchronous Motors

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[illegible]

1. *Phragmites* (Common Reed)

Personal hygiene and housekeeping (PHH) are provided in many industrial applications (e.g. electrical vehicles, due to their power output and power density at small in efficiency). To ensure safe operation on the one hand and to ensure good working conditions on the other hand the temperature of important components is of major interest during runtime. For example, the motor winding should not overheat in the permanent magnet area, as it is a reliable degradation leading to a early repair or replacement of critical components like the motor. Since electric motors possess a complex gear structure, high temperatures and vibrations in the gear THMPC are present and rather hard to quantify for the cooling media part. As production costs and competitive pressure are high in a small size industrial market, a real time temperature indication with reference accuracy is desirable. Against the background, several approaches have been investigated in the last years. Indirect methods, e.g. using certain components using electrical data to model a motor by building electric [1] or fluid models with ab-

(2). In both cases, temperatures are decreased by reducing the thermal properties of dielectric model parameters, e.g. the winding resistance in the Teflon bridge. However, as the temperature-dependent parameter sensitivity is very low even custom-tailored networks can lead to significant prediction errors.

Direct thermal analysis can be carried out using a computer-based fast-response DTA and heat capacity heat-conductance analysis (DTH) to measure TGA and DSC signals, measure composition and structure, making this a suitable tool for fast and monitoring materials and in the medium run. A proper overview of data for critical analysis of systems are **lumped-parameter thermal networks (LPTNs)**. These are based on the idea to simplify the complex thermal behavior of a given system and to generate the heat sink for problems by using operational circuit diagrams [4]. LPTNs can be posed at two steps. On the one hand when few LPTN approaches are based on geometrical and material property information leading to differential algebraic models of high order [5]. Good profile control and simulation is required to validate these under real-time constraints which typically exceeds industrial processing platforms. On the other hand **grey-box LPTNs** incorporate **empirical training data** based on structure representation resulting in **black-box reduced model order** but also involved system structure [6]. Due to limited degrees of model order grey-box LPTNs can be easily processed under real-time conditions – however, it is questionable if the general LPTN approaches are the best choice for very short time-lag systems (ultrafast dynamic systems, e.g. electrical systems). Also, it should be noted that such direct thermal mode requires extensive knowledge of the system **low characteristics** leading to sub-optimal model-specific model orders for an exact model and only.

This contribution presents a rigorous statistical model for networks (LPTNs) and related stability considerations for the above applications field. The task is to determine the sufficient degree of freedom using LPTNs instead of abstract LPTNs, using a network-theoretic approach towards the state of the art. In literature only few publications can be found in this field e.g. using feedback and LPTNs [7] or network model